



CALIBER 2026 — CASE 2: INTELLIGENT MANUFACTURING

One file. One truth. From data to decision.

380 incidents over 31 months, read from a single governed table. No server, no internet, no install.

\$67.19M

USD exposure in scope

\$2,261 h

downtime across 380 records

179

overdue actions (109 past 1 year)

87.1%

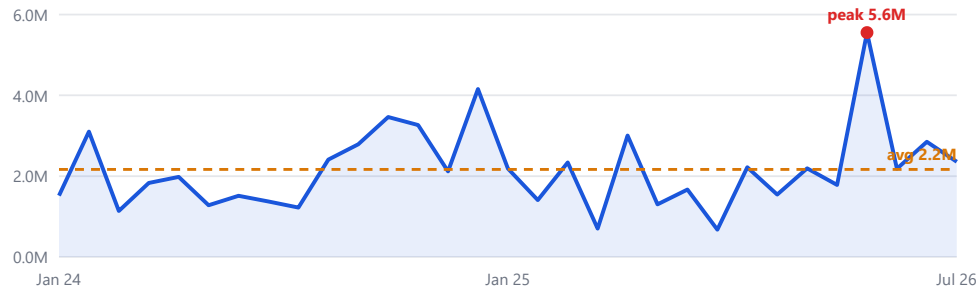
AI retrieval @ rank 1



Monthly loss exposure, Jan 2024 → Jul 2026 — peak \$5.6M, average \$2.17M per month.

The loss curve is going up, not down.

MONTHLY LOSS EXPOSURE — 31 MONTHS



+34%

2026 run-rate vs 2024 (\$1.97M → \$2.64M/mo)

\$5.56M

worst single month (Apr 2026)

\$56%

of exposure sits behind an overdue action

109

actions already overdue by more than a year

- Six functions, six dashboards, six KPI definitions — time spent validating before deciding.
- Alerts fire, but nobody owns them: 220 actions open, 179 past due.
- Root cause found late, so the same mechanism keeps repeating.

What we inherited — measured, not asserted.

EXPOSURE BY FAILURE MECHANISM

#	MECHANISM	CASES	LOSS	SHARE	
1	Leakage	101	\$17.00M	25.3%	
2	High Vibration	57	\$9.88M	14.7%	
3	Worn Out	47	\$8.17M	12.2%	
4	Crack	20	\$6.55M	9.8%	
5	Fouling	20	\$3.70M	5.5%	

Top 3 mechanisms = 52% of total exposure (\$35.04M of \$67.19M) in just 205 of 380 records.

WHERE THE MONEY IS, BY UNIT

#	UNIT	CASES	OVERDUE	LOSS	
1	ZCU	62	29	\$11.11M	
2	ARP	49	26	\$9.08M	
3	OPP	42	22	\$7.54M	
4	SMX	42	21	\$7.43M	
5	OP2	37	15	\$5.81M	
6	NUP	24	10	\$5.49M	

Unit ZCU alone carries \$11.11M and 29 overdue actions.

380

incidents / 23 columns

5

assets x 26 weeks condition

5 x 721

hourly PI production rows

5

structured RCA decks (4P / 4M+1E)

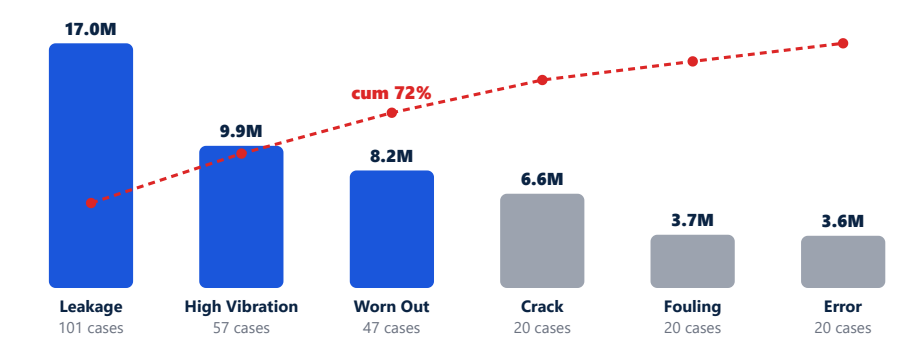
Four sources in. One governed table out.

Every chart on every tab reads the same incident table. Edit one record, every figure recomputes.



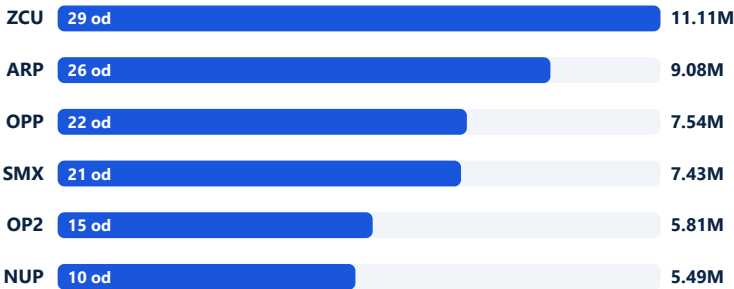
The executive view, in one screen.

FIX-FIRST PRIORITY — PARETO OF LOSS



Blue = top 3. Red dashes = cumulative share. Tapping a bar filters the incident list to that mechanism.

UNIT RANKING — LOSS WITH OVERDUE COUNT



Bar = loss in USD. **od** = overdue actions behind that unit. Tap to filter.

\$67.19M

hero exposure card

179

overdue, owner + age on every row

11

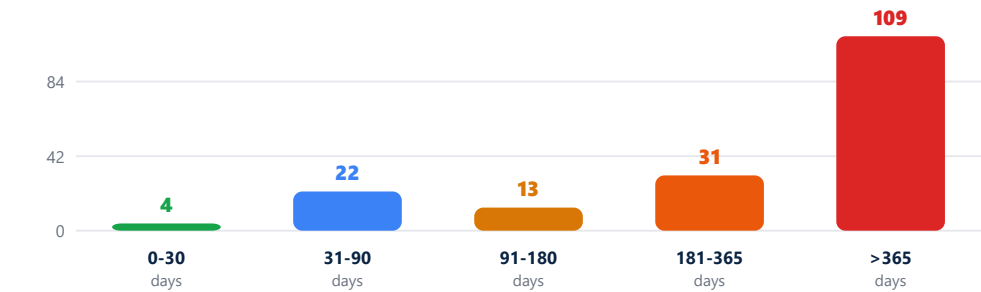
views, every KPI clickable to source

0

charts without a data provenance link

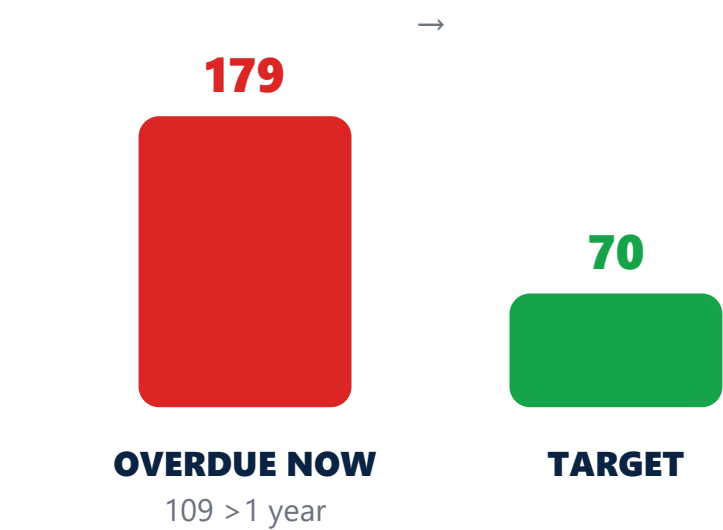
109 actions are more than a year late.

OVERDUE ACTIONS BY AGE (179 TOTAL)



Real distribution of 179 overdue actions measured at 19-Aug-2026. Median age 463 days, oldest 926 days.

BEFORE VS TARGET

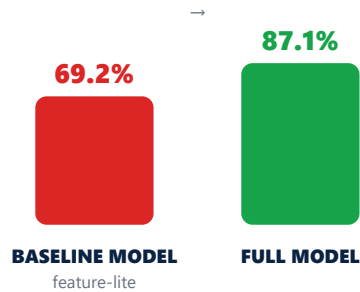


Target basis: every action gets an owner and a due date at creation, so nothing can age past its due date — the board blocks closure without evidence.

- Red dot = nobody on it. Purple = in progress. Green = done.
- Overdue carries \$37.32M of exposure (56% of total).

AI suggests. Humans close.

SIMILAR-INCIDENT RETRIEVAL, BEFORE VS AFTER



- **87.1%** correct match at rank 1 (full model)
- **98.4%** at rank 5
- Baseline model with thin features: **69.2%**
- **+17.9 points** from uniting the four sources

Both numbers measured leave-one-out on all 380 records — not a demo set. Full model 95% CI 83.4–90.1%.

87.1%

rank-1 retrieval, CI 83.4–90.1%

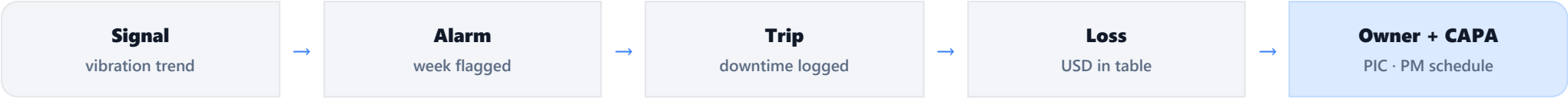
+17.9 pts

vs feature-lite baseline (69.2%)

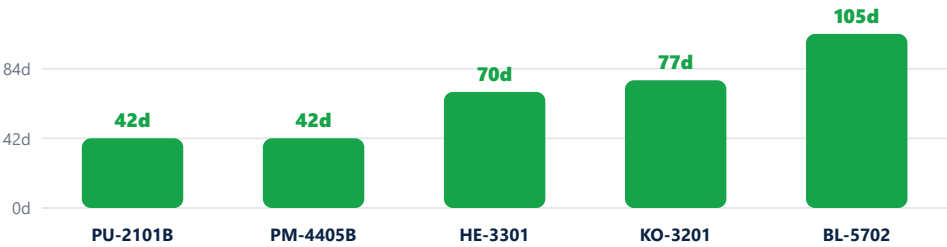
- Problem tank → root-cause indication → prioritized alerts → tracked action.
- **Top-10 exposure plan** drafted in one click, each row with owner + PM schedule.
- Human-in-the-loop: AI never closes an action, the PIC does.

Full method, confidence intervals and honest limits published in the Model Evaluation tab.

From signal to owner, and 67 days of warning.



WARNING WINDOW PER ASSET — FIRST ALARM → TRIP



Real gap from 26-week condition history. Mean **67 days**, range 42–105 days — the time available to act before a trip.

WHY IT MATTERS

ASSET	LEAD	READ
PU-2101B	42 d	Actionable
PM-4405B	42 d	Actionable
HE-3301	70 d	Before trip
KO-3201	77 d	Before trip
BL-5702	105 d	Before trip

All 5 RCA decks integrated (4P / 4M+1E, CAPA, PIC, status, PM schedule) — chronology and correlation one tap from any row.

Four numbers we commit to moving.

Baseline column = measured from the committee data. Target column = modeled commitment with its basis stated.

METRIC	BEFORE		TARGET		BASIS
Monthly loss	\$2.64M/mo	→	\$1.97M/mo	-25%	2026 running rate vs 2024 level. Fix-first on top-3 mechanisms covers 52% of exposure.
Overdue actions	179 of 220	→	70	-61%	109 actions already past due by 1 year+. Board forces an owner and a due date at creation.
Retrieval @1	69.2%	→	87.1%	+17.9 pts	Both measured leave-one-out on n=380. Full model CI 83.4-90.1%.
Warning window	reactive	→	67 days	new	Mean gap between first ALARM week and TRIP week across all 5 assets, from 26-week condition history.

Targets are modeled from this baseline, not measured results. We publish the basis so the assumption can be challenged, not the number.

What we measure, what we label, what we skip.

MEASURED

- Loss, downtime, overdue aging — from 380 real records
- Retrieval 87.1% / 98.4% leave-one-out
- Warning window 67 days from condition history

LABELED, NOT HIDDEN

- Energy = derived proxy, formula printed on the slide
- CCTV wall = illustration, unit aggregates underneath are real
- Baseline defects listed in Data Governance

OUT OF SCOPE

- No live streaming (not required) — ready for PI stream
- No automated closure — human-in-the-loop by design
- No extra data beyond two declared additions

52%

of exposure fixable by top-3 focus

\$35.04M

addressable in 3 mechanisms

109

stale actions surfaced with age

67 d

average early-warning window

Jenny · Hafidz · Naufal

Advisor: Mr. Bayu Aji. Built from the official baseline only, plus two declared additions.

1 — Meter it

5 metered energy tags to replace the proxy formula

2 — Connect it

EDMS / AIMS / Digital Twin for asset context

3 — Stream it

PI live feed, architecture already decoupled for it

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